Homogeneous Function:

A function $u = f(x, y)$ is said to be homogeneous function of degree n, if

$u = x^n f\left(\frac{y}{x}\right)$, where n is a real number.

Note:- Degree of a homogeneous function $u = f(x,$ can be obtained by replacing x by tx a y by ty, if

$$f(tx, ty) = t^n f(x, y) = t^n \cdot u$$

Then u is a homogeneous function of degree n Same method can be extended for a function o more than two variables.

e.g. (1) $u = \frac{x^2 + y^2}{x + y} = \frac{x^2\left(1 + \frac{y^2}{x^2}\right)}{x\left(1 + \frac{y}{x}\right)} = x \cdot \left[\frac{1 + \frac{y^2}{x^2}}{1 + \frac{y}{x}}\right]$

$= x \cdot f\left(\frac{y}{x}\right)$

$n = 1 \Rightarrow$ Degree is 1.

(2) $u = x^2 + 2xy + y^2$

$u = f(x, y) = x^2 + 2xy + y^2$

$f(tx, ty) = (tx)^2 + 2(tx)(ty) + (ty)^2$

$= t^2 [x^2 + 2xy + y^2]$

$= t^n f(x, y)$

$n = 2 \Rightarrow$ Degree is 2.

e.g. (3) $u = f(x, y) = x^{\frac{3}{2}} y^2 + x^{-\frac{3}{2}} y^2$

Here u is not a homogeneous function because degree of each term is not same.

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Euler's thm. for function of two variables for homogeneous functions:

Theorem:

If u is a homogeneous function of two variables x and y of degree n then

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu.$$

Proof:-

since, u is a homogeneous function of degree n in two variables x & y,

$\therefore$ By def^{n}, $u = x^n f(y/x)$ — (1)

Diff. eq^{n} (1) partially w.r. to x,

$$\frac{\partial u}{\partial x} = n \cdot x^{n-1} f(y/x) + x^n f'(y/x)\left(\frac{-y}{x^2}\right)$$

$$= n \cdot x^{n-1} f(y/x) - x^{n-2} y f'(y/x) \quad — (2)$$

Again, diff., eq^{n} (1) partially w.r. to y,

$$\frac{\partial u}{\partial y} = x^n \cdot f'(y/x) \cdot 1/x$$

$$= x^{n-1} f'(y/x) \quad — (3)$$

Now, ~~From~~ (2),

$$x\frac{\partial u}{\partial x} = n \cdot x^n f(y/x) - x^{n-1} y f'(y/x) \quad — (4)$$

From (3),

$$y\frac{\partial u}{\partial y} = x^{n-1} y f'(y/x) \quad — (5)$$

Adding eq^{ns} (4) & (5),

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n \cdot x^n f(y/x) - x^{n-1} y f'(y/x) + x^{n-1} y f'(y/x)$$

$$= n \cdot x^n f(y/x)$$

$$= n \cdot u$$

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Note :-

In general, if u is a homogeneous function in degree n in the n-variables $x_1, x_2, \ldots, x_n$ then

$$x_1 \frac{\partial u}{\partial x_1} + x_2 \frac{\partial u}{\partial x_2} + \cdots + x_n \frac{\partial u}{\partial x_n} = nu.$$

Classroom problems:

Ex: 1 Verify Euler's thm. for i) $u = ax^2 + 2hxy + by^2$

ii) $u = \sqrt{x} + \sqrt{y} + \sqrt{z}$.

i) $u = ax^2 + 2hxy + by^2$.

Replacing x by xt, y by yt,

$$u = f(x,y) = f(tx, ty) = a(tx)^2 + 2h(tx)(ty) + b(ty)^2$$
$$= t^2[ax^2 + 2hxy + by^2]$$
$$= t^2 f(x,y)$$

$\therefore u$ is a homogeneous function of degree 2.

$\therefore$ By Euler's thm,

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$$
$$= 2u. \quad \text{— (1)}$$

To verify Euler's thm, diff. u partially w.r to x,

$$\frac{\partial u}{\partial x} = 2ax + 2hy \quad , \quad \frac{\partial u}{\partial y} = 2hx + 2by$$

$$x \frac{\partial u}{\partial x} = 2ax^2 + 2hxy \quad , \quad y \frac{\partial u}{\partial y} = 2hxy + 2by^2$$

Now, $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2ax^2 + 2hxy + 2hxy + 2by^2$

$$= 2[ax^2 + 2hxy + by^2]$$
$$= 2u. \quad \text{— (2)}$$

From eq^{n} (1) & (2), thm. is verified.

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ii) $u = \sqrt{x} + \sqrt{y} + \sqrt{z}$.

Replacing x by tx, y by ty

$u = f(x,y) = f(tx, ty) = \sqrt{tx} + \sqrt{ty} + \sqrt{tz} = t^{1/2}(\sqrt{x} + \sqrt{y} + \sqrt{z})$

$\therefore u$ is a homogeneous function of degree $1/2$.

$\therefore$ By Euler's thm,

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{1}{2}u. \quad \text{——(1)}$$

To verify Euler's thm.,

$$\frac{\partial u}{\partial x} = \frac{1}{2\sqrt{x}}, \quad \frac{\partial u}{\partial y} = \frac{1}{2\sqrt{y}}, \quad \frac{\partial u}{\partial z} = \frac{1}{2\sqrt{z}}$$

$$x\frac{\partial u}{\partial x} = \frac{\sqrt{x}}{2}, \quad y\frac{\partial u}{\partial y} = \frac{\sqrt{y}}{2}, \quad z\frac{\partial u}{\partial z} = \frac{\sqrt{z}}{2}.$$

Now, $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = \frac{\sqrt{x}}{2} + \frac{\sqrt{y}}{2} + \frac{\sqrt{z}}{2} = \frac{1}{2}u$. —(2)

$\therefore$ From (1) & (2), Euler's thm. is varified.

2 If $u(x,y) = \frac{1}{x^2} + \frac{1}{xy} + \frac{\log x - \log y}{x^2}$, prove that

$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + 2u(x,y) = 0$.

Here, $u(x,y) = \frac{1}{x^2} + \frac{1}{xy} + \frac{\log x - \log y}{x^2}$

$$= \frac{1}{x^2} + \frac{1}{xy} + \frac{1}{x^2}\log\left(\frac{x}{y}\right)$$

Replacing x by xt, y by yt,

$$u(x,y) = \frac{1}{t^2x^2} + \frac{1}{txty} + \frac{1}{t^2x^2}\log\left(\frac{x}{y}\right)$$

$$= t^{-2}\left[\frac{1}{x^2} + \frac{1}{xy} + \frac{1}{x^2}\log\left(\frac{x}{y}\right)\right]$$

Hence, u is a homogeneous f^{n} of degree -2.

$\therefore$ By Euler's thm., $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = -2u(x,y)$

$\therefore x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + 2u(x,y) = 0$

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Ex:3 If $u = \frac{x^2y^2z^2}{x^2+y^2+z^2} + \cos\left(\frac{xy+yz}{x^2+y^2+z^2}\right)$ then

show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = \frac{4x^2y^2z^2}{x^2+y^2+z^2}$.

Let, $u = v + w$

where $v = \frac{x^2y^2z^2}{x^2+y^2+z^2}$, $w = \cos\left(\frac{xy+yz}{x^2+y^2+z^2}\right)$

Replacing x by tx, y by ty, z by tz

$v = t^4\left(\frac{x^2y^2z^2}{x^2+y^2+z^2}\right)$, $w = t^0 \cos\left(\frac{xy+yz}{x^2+y^2+z^2}\right)$.

$\therefore$ v is a homogeneous f^n of degree 4 and w is a homogeneous f^n of degree 0.

$\therefore$ By Euler's thm.,

$x\frac{\partial v}{\partial x} + y\frac{\partial v}{\partial y} + z\frac{\partial v}{\partial z} = 4v$ —— (1)

$x\frac{\partial w}{\partial x} + y\frac{\partial w}{\partial y} + z\frac{\partial w}{\partial z} = 0 \cdot w = 0$ —— (2)

(1) + (2)

$x\left(\frac{\partial v}{\partial x} + \frac{\partial w}{\partial x}\right) + y\left(\frac{\partial v}{\partial y} + \frac{\partial w}{\partial y}\right) + z\left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial z}\right) = 4v$

$\therefore$ $x\frac{\partial(v+w)}{\partial x} + y\frac{\partial(v+w)}{\partial y} + z\frac{\partial(v+w)}{\partial z} = 4v$

$\therefore$ $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = \frac{4x^2y^2z^2}{x^2+y^2+z^2}$.

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Deductions from Euler's thm.

Corollary 1:

If u is a homogeneous function of two variables x & y of degree n then

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = n(n-1)u.$$

Let, u be a homogeneous fⁿ of two variables x & y of degree n.

∴ By Euler's thm.,

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu \quad \text{—— (1)}$$

Differentiating eqⁿ (1) partially w.r.to x,

$$\frac{\partial u}{\partial x} + x\frac{\partial^2 u}{\partial x^2} + y\frac{\partial^2 u}{\partial x \partial y} = n\frac{\partial u}{\partial x}.$$

$$\therefore x\frac{\partial^2 u}{\partial x^2} + y\frac{\partial^2 u}{\partial x \partial y} = (n-1)\frac{\partial u}{\partial x}. \quad \text{—— (2)}$$

Differentiating eqⁿ (1) partially w.r.to y,

$$x\frac{\partial^2 u}{\partial x \partial y} + \frac{\partial u}{\partial y} + y\frac{\partial^2 u}{\partial y^2} = n\frac{\partial u}{\partial y}$$

$$\therefore x\frac{\partial^2 u}{\partial x \partial y} + y\frac{\partial^2 u}{\partial y^2} = (n-1)\frac{\partial u}{\partial y} \quad \text{—— (3)}$$

Multiplying eqⁿ (2) by x, (3) by y and then adding,

$$x^2\frac{\partial^2 u}{\partial x^2} + xy\frac{\partial^2 u}{\partial x \partial y} + xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = (n-1)x\frac{\partial u}{\partial x} + (n-1)y\frac{\partial u}{\partial y}$$

$$\therefore x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = n(n-1)\left[x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}\right]$$

$$= n(n-1)u.$$

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Homogeneous

Class room problems!

① If $u = x \cdot f\left(\frac{y}{x}\right) + g\left(\frac{y}{x}\right)$, show that $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 0$.

Let, $u = v + w$.

Here, $v = x \cdot f\left(\frac{y}{x}\right)$, $w = g\left(\frac{y}{x}\right)$.

Replacing x by tx, y by ty,

$v = tx \cdot f\left(\frac{y}{x}\right)$, $w = t^0 g\left(\frac{y}{x}\right)$

∴ v is a homogeneous f^n of degree 1 and w is a homogeneous f^n of degree 0.

∴ By Corollary 1,

$$x^2 \frac{\partial^2 v}{\partial x^2} + 2xy \frac{\partial^2 v}{\partial x \partial y} + y^2 \frac{\partial^2 v}{\partial y^2} = n(n-1)v = 0 \quad \text{—} (1)$$

$$y^2 \frac{\partial^2 w}{\partial x^2} + 2xy \frac{\partial^2 w}{\partial x \partial y} + y^2 \frac{\partial^2 w}{\partial y^2} = n(n-1)w = 0 \quad \text{—} (2)$$

∴ (1) + (2),

$$x^2 \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 w}{\partial x^2}\right) + 2xy \left(\frac{\partial^2 v}{\partial x \partial y} + \frac{\partial^2 w}{\partial x \partial y}\right) + y^2 \left(\frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 w}{\partial y^2}\right) = 0$$

$$\therefore x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 0.$$

② If $u = \frac{x^3+y^3}{y\sqrt{x}} + \frac{1}{x^7} \sin^{-1}\left(\frac{x^2+y^2}{x^2+2xy}\right)$, find value of

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} + x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \text{ at } x=1, y=2.$$

Let, $u = v + w$ where,

$v = \frac{x^3+y^3}{y\sqrt{x}}$, $w = \frac{1}{x^7} \sin^{-1}\left(\frac{x^2+y^2}{x^2+2xy}\right)$

Replacing x by tx, y by ty,

$$v = \frac{t^3(x^3+y^3)}{t^{3/2}\, y\sqrt{x}} = \frac{t^{3/2}(x^3+y^3)}{y\sqrt{x}}.$$

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$$w = t^{-7}\left[\frac{1}{x^7}\sin^{-1}\left(\frac{x^2+y^2}{x^2+2xy}\right)\right]$$

$\therefore$ u is a homogeneous f^{n} of degree $\frac{7}{2}$ and w is a homogeneous f^{n} of degree -7.

$\therefore$ By Euler's thm. and corollary 1,

$$x^2\frac{\partial^2 v}{\partial x^2} + 2xy\frac{\partial^2 v}{\partial x\partial y} + y^2\frac{\partial^2 v}{\partial y^2} + x\frac{\partial v}{\partial x} + y\frac{\partial v}{\partial y} = \frac{3}{2}\left(\frac{3}{2}-1\right)v + \frac{3}{2}v$$

$$= \frac{9}{4} - \frac{3}{2} + \frac{3}{2}v$$

$$= \frac{9}{4}v \quad \text{—— (1)}$$

$$x^2\frac{\partial^2 w}{\partial x^2} + 2xy\frac{\partial^2 w}{\partial x\partial y} + y^2\frac{\partial^2 w}{\partial y^2} + x\frac{\partial w}{\partial x} + y\frac{\partial w}{\partial y} = -7(-7-1)w + (-7)v$$

$$= -7(-8)w - 7v$$

$$= 49w \quad \text{—— (2)}$$

Now (1) + (2),

$$x^2\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 w}{\partial x^2}\right) + 2xy\left(\frac{\partial^2 v}{\partial x\partial y} + \frac{\partial^2 w}{\partial x\partial y}\right) + y^2\left(\frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 w}{\partial y^2}\right) +$$

$$x\left(\frac{\partial v}{\partial x} + \frac{\partial w}{\partial x}\right) + y\left(\frac{\partial v}{\partial y} + \frac{\partial w}{\partial y}\right) = \frac{9}{4}v + 49w$$

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} + x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{9}{4}v + 49w$$

At $x=1$, $y=2$,

$$v = \frac{1^3+2^3}{2\sqrt{1}} = \frac{9}{2}, \quad w = \frac{1}{1}\sin^{-1}\left(\frac{1^2+2^2}{1^2+2\cdot1\cdot2}\right) = 1\cdot\sin^{-1}(1)$$

and $\sin^{-1}(1) = \frac{\pi}{2}$.

$$\therefore x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} + x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{9}{4}\left(\frac{9}{2}\right) +$$

$$+ 49\cdot\frac{\pi}{2}$$

$$= \frac{81}{8} + \frac{49\pi}{2}$$

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Modified Euler's thm:-

If $z = f(u)$ is a homogeneous function of degree n in variables x and y then

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\frac{f(u)}{f'(u)}$$

Proof

By Euler's thm.,

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = nz = n f(u)$$

$$x\cdot\frac{\partial}{\partial x}f(u) + y\frac{\partial}{\partial y}f(u) = n\cdot f(u)$$

$$x\cdot f'(u)\frac{\partial u}{\partial x} + y\cdot f'(u)\frac{\partial u}{\partial y} = n f(u)$$

$$\therefore \left[x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}\right] f'(u) = n f(u)$$

$$\therefore x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\cdot\frac{f(u)}{f'(u)}.$$

Note:-

If $v = f(u)$ is a homogeneous function of degree n in variables x, y and z then

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = n\cdot\frac{f(u)}{f'(u)}$$

(Before starting corollary : 2, examples of modified Euler thm should be done).

Corollary 2:-

If $z = f(u)$ is a homogeneous function of degree n in variables x and y then

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = g(u)\,[g'(u) - 1],$$ where $g(u) = n$

Proof:-

By Modified Euler's thm.,

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\cdot\frac{f(u)}{f'(u)} = g(u) \quad \text{---} \quad (1)$$

Diff. eq^n (1) partially w.r.t. to x,

$$x\frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x} + y\frac{\partial^2 u}{\partial x\partial y} = g'(u)\frac{\partial u}{\partial x}.$$

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$$\therefore x\frac{\partial^2 u}{\partial x^2} + y\frac{\partial^2 u}{\partial x \partial y} = g'(u)\frac{\partial u}{\partial x} - \frac{\partial u}{\partial x} = \frac{\partial u}{\partial x}[g'(u)-1] \quad \text{— (2)}$$

Diff. eq^n (1) partially w.r. to y,

$$x\frac{\partial^2 u}{\partial x \partial y} + y\frac{\partial^2 u}{\partial y^2} + \frac{\partial u}{\partial y} = g'(u)\frac{\partial u}{\partial y}$$

$$\therefore x\frac{\partial^2 u}{\partial x \partial y} + y\frac{\partial^2 u}{\partial y^2} = g'(u)\frac{\partial u}{\partial y} - \frac{\partial u}{\partial y} = \frac{\partial u}{\partial y}[g'(u)-1] \quad \text{— (3)}$$

Multiplying eq^n (2) by x, (3) by y and adding,

$$x^2\frac{\partial^2 u}{\partial x^2} + xy\frac{\partial^2 u}{\partial x \partial y} = [g'(u)-1]\, x\cdot\frac{\partial u}{\partial x}$$

$$y^2\frac{\partial^2 u}{\partial y^2} + xy\frac{\partial^2 u}{\partial x \partial y} = [g'(u)-1]\, y\frac{\partial u}{\partial y}$$

$$\therefore x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = [g'(u)-1]\left[x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}\right]$$

$$= [g'(u)-1]\, g(u) \quad \{\text{from (1)}$$

where $g(u) = n\dfrac{f(u)}{f'(u)}$.

Class room problems:

If $u = \sec^{-1}\left(\dfrac{x^3+y^3}{x+y}\right)$, prove that $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = 2\cot u$.

Here, $u = \sec^{-1}\left(\dfrac{x^3+y^3}{x+y}\right)$

Replacing x by tx, y by ty,

$$u = \sec^{-1}\left[t^2\left(\frac{x^3+y^3}{x+y}\right)\right]$$

u is a non-homogeneous function ~~of degree 2~~

But $\sec u = \dfrac{x^3+y^3}{x+y}$ is a homogeneous function of degree 2.

Let, $f(u) = \sec u$.

$\therefore$ By modified Euler's thm.,

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\cdot\frac{f(u)}{f'(u)} = 2\cdot\frac{\sec u}{\sec u\tan u} = 2\cot u$$

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homogeneous

② If $u = \log\left(\frac{x^7+y^7+z^7}{x+y+z}\right)$, s.t. $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 6$.

Here, $u = \log\left(\frac{x^7+y^7+z^7}{x+y+z}\right)$

Replacing x by tx, y by ty,

$$u = \log\left[\frac{t^7(x^7+y^7+z^7)}{t(x+y+z)}\right] = \log\left[t^6\frac{(x^7+y^7+z^7)}{x+y+z}\right]$$

u is a non-homogeneous function but

$e^u = \frac{x^7+y^7+z^6}{x+y+z}$ is a homogeneous f^n of degree 6

Let $f(u) = e^u$.

∴ By Modified Euler's thm.,

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 6\cdot\frac{e^u}{e^u} = 6.$$

③ If $u = \operatorname{cosec}^{-1}\left(\sqrt{\frac{x^{1/2}+y^{1/2}}{x^{1/3}+y^{1/3}}}\right)$, prove that

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = \frac{\tan u}{144}(13+\tan^2 u)$$

Here, $u = \operatorname{cosec}^{-1}\left(\sqrt{\frac{x^{1/2}+y^{1/2}}{x^{1/3}+y^{1/3}}}\right)$

$$= \operatorname{cosec}^{-1}\left(\sqrt{\frac{t^{1/2}(x^{1/2}+y^{1/2})}{t^{1/3}(x^{1/3}+y^{1/3})}}\right)$$

$$= \operatorname{cosec}^{-1}\left(\sqrt{t^{1/6}\frac{(x^{1/2}+y^{1/2})}{(x^{1/3}+y^{1/3})}}\right)$$

$$= \operatorname{cosec}^{-1}\left[t^{1/2}\sqrt{\frac{x^{1/2}+y^{1/2}}{x^{1/3}+y^{1/3}}}\right].$$

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~~u is a homogeneous fn of degree $\frac{1}{12}$.~~

u is a non-homogeneous function but $\operatorname{cosec} u = \sqrt{\frac{x^{1/2}+y^{1/2}}{x^{1/3}+y^{1/3}}}$ is a homogeneous fn of degree $\frac{1}{12}$.

Let, $f(u) = \operatorname{cosec} u$.

$\therefore$ By Corollary 2,

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = g(u)[g'(u)-1]$$

where $g(u) = \frac{n \cdot f(u)}{f'(u)}$

$\therefore g(u) = \frac{1}{12}\frac{\operatorname{cosec} u}{(-\operatorname{cosec} u \cdot \cot u)} = -\frac{1}{12}\tan u$.

$g'(u) = -\frac{1}{12}\sec^2 u = -\frac{1}{12}(1+\tan^2 u)$

$\therefore x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = -\frac{1}{12}\tan u\left[-\frac{1}{12}(1+\tan^2 u)-1\right]$

$= -\frac{1}{12}\tan u\left[\frac{-1-\tan^2 u-12}{12}\right]$

$= +\frac{1}{144}\tan u\,(13+\tan^2 u)$

If $u = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$, prove that

i) $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \sin 2u$

ii) $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = 2\sin u \cdot \cos 3u$.

Here, $u = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$

$\therefore u = \tan^{-1}\left[t^2\left(\frac{x^3+y^3}{x+y}\right)\right]$

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Given -

u is a non-homogeneous ... but

$\tan u = \frac{x^3+y^3}{x-y}$ is a homogeneous function of degree 2.

Let, $f(u) = \tan u$.

$\therefore$ By Modified Euler's thm.,

$$x\frac{\partial u}{\partial x} + y\frac{\partial y}{\partial y} = n\cdot\frac{f(u)}{f'(u)}$$

$$= 2\cdot\frac{\tan u}{\sec^2 u}$$

$$= 2\cdot\sin u\cdot\cos u$$

$$= \sin 2u.$$

Now, By corollary 2,

$$x^2\frac{\partial^2 y}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 y}{\partial y^2} = g(u)[g'(u)-1]$$ where

$$g(u) = n\frac{f(u)}{f'(u)}$$

$$\therefore g(u) = 2\frac{\tan u}{\sec^2 u} = \sin 2u.$$

$$g'(u) = 2\cos 2u$$

$$\therefore x^2\frac{\partial^2 y}{\partial x^2} + 2xy\frac{\partial^2 y}{\partial x\partial y} + y^2\frac{\partial^2 y}{\partial y^2} = \sin 2u\,[2\cos 2u - 1]$$

$$= 2\cdot\sin 2u\cdot\cos 2u - \sin 2u$$

$$= \sin 4u - \sin 2u$$

$$= 2\sin u\cdot\cos 3u$$

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